MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, CCU-28/A ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Cartridge, Impulse, CCU-28/A, Assembly, for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-E-5272	- Environmental Testing, Aeronautical, and Associated Equipment, General Specification For
MIL-C-60539	- Coating, Anodic, Conventional, for Aluminum and Aluminum Alloys.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings
MS 28741	- Hose Assembly - Detachable End Fitting, Medium Pressure

DRAWINGS

Armament Command

F11751078	- Cartridge, Impulse, CCU-28/A Assembly
D9282893	- Box, Fiberboard (Inner) Packing Ammunition for Cartridge, Impulse CCU-28/A
D9282894	- Box, Packing, Ammunition For Cartridge, Impulse, CCU-28/A
D9282895	- Box, Fiberboard (Outer) Alternative Packing, Ammunition, for Cartridge Impulse, CCU-28/A
IEL-11751078	- Inspection Equipment List for Cartridge, Impulse, CCU-28/A

FSC 1377

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2.1 (cont'd)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following documents forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018).

3. REQUIREMENTS

3.1 General.-Materials, parts, and assemblies shall conform to the applicable drawings and referenced specification.

3.2 Protective finish.-Protective finish shall be in accordance with Specification MIL-C-60539.

3.3 Vibration.-The assembled cartridges shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant or derangement of components.

3.4 Functional requirements.

3.4.1 The cartridge shall not fail to fire.

3.4.2 Functioning.-The cartridge when assembled and tested in an acceptable RAU-1/A Remover shall propel a 362.5 pound mass vertically and shall comply with the requirements of Table I.

TABLE I

Temperature Degrees (°) (Fahrenheit)	Minimum Velocity Feet Per Second (ft/sec)	Maximum Thrust (Pounds)	Ignition Delay (Milliseconds)
-65° to -70°	25	5400	40 max.
65° to 75°	25	5600	40 max.
200° to 205	25	6100	40 max.

3.4.2.1 Initiation.-The remover shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 10 feet of Aircraft Hose, Size No. 4, in accordance with MS28741 at $70^{\circ} \pm 5^{\circ}$F.

3.5 Workmanship.-Workmanship shall be in accordance with the requirements of the applicable drawings, specifications and the following:

3.5.1 Metal defects.-The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

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3.5.2 Burr.-The part shall not have a burr whixh might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.-The part or assembly shall not contain chips.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilitities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to defined the quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 5 cartridges for sealant tests, 21 cartridges for ballistic tests, 12 for vibration tests (to be selected from ballistic sample) and 10 sets of unassembled components for dimensional and visual examination and tests.

The sample shall be manufactured using the same material, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specification shall result in sample disapproval.

TABLE I

1 - 300	100% Examination	-	-	-	
301 - 800	50	0	1	1	2
801 - 1800	50	1	2	2	3

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4.3 Inspection provisions

4.3.1 Lot

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.An ammunition lot shall consist of an identifiable group of metal parts assemblies consisting of:

a. Components produced by a homogeneous process
b. Propellant from not more than one lot.
c. Primers from not more than one lot.
d. Igniter from not more than one lot.

4.3.2 Examination.-One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables II and III. For an individual defect (single characteristic), the criteria in Table II shall be used. For total defects within a major or minor class, the criteria in Table III shall be used. All non-conforming material shall be rejected. The roughness comparison specimens prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examinationfor such defects as burr or foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE III

Number of defect in class	Total Allowable Defects Per Sample Size Sample Size for Major Defects			
	110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 19	4	5	8	10
20 - 43	6	8	11	14
over 43	8	11	17	20

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4.3.2.1 Firing Plug Drawing B8593498, covering a detail of Drawing D11751078.

Defect	Method of Inspection
CRITICAL	
1. Length of striker tip, min.	Gage
MAJOR	
101. Diameter of body, min.	Gage
102. Profile of striker tip	Gage
103. Height of boss	Gage
104. Metal defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Diameter of body, max.	Gage
202. Diameter of boss	Gage
203. Total length	Gage
204. Length of striker tip, max.	Gage
205. Surface finish improper	Visual

4.3.2.2 Ring, Retaining Dwg B8593499, covering a detail of Dwg. F11751078.

CRITICAL	
None defined	
MAJOR	
101. Concentricity of center hole with body	Gage
102. Diameter of body, min.	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr	Visual
MINOR	
201. Diameter of center hole	Gage
202. Thickness of body, max.	Gage

4.3.2.3 Case, Cartridge, Drawing C8593736 covering a detail of Drawing F11751078.

CRITICAL	
None defined	

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4.3.2.3 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Concentricity of head with body	Gage
102. Straightness of base	Gage
103. Diameter of body, min.	Gage
104. Diameter of head, min.	Gage
105. Thickness of base, min.	Gage
106. Thickness of sidewall, min.	Gage
107. Protective finish inadequate or defective	Visual
108. Burr or sharp corner	Visual
109. Foreign matter	Visual
110. Metal defective	Visual
MINOR	
201. Total length	Gage
202. Thickness of sidewall max.	Gage
203. Diameter of body, max.	Gage
204. Diameter of head max.	Gage
205. Location of datum diameter	Gage
206. Radius or chamfer missing or incorrect	Visual
207. Surface finish improper	Visual

4.3.2.4 Disc, Sealing, Drawing C8593494 covering a detail of Drawing F11751078.

CRITICAL	
None defined	
MAJOR	
101. Diameter, min.	Gage
102. Marking, missing, incorrect or illegible	Visual
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Diameter, max.	Gage
202. Thickness	Gage
203. Varnish missing	Visual

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4.3.2.5 Igniting Disc, Drawing B8593355 covering a detail of Drawing F11751078.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter, min.	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Diameter, max.	Gage
202. Thickness	Gage

4.3.2.6 Head, Cartridge, Drawing C11731601 covering a detail of Drawing F11751078.

CRITICAL	
1. Thickness of metallic web between flash hole and igniter cavity	Gage
MAJOR	
101. Concentricity of flange with body	Gage
102. Concentricity of counterbore with firing pin plug cavity	Gage
103. Concentricity of firing plug cavity with body	Gage
104. Concentricity of firing plug cavity with primer percussion cavity	Gage
105. Diameter of counterbore at flange end, max.	Gage
106. Wall thickness min.	Gage
107. Surface finish firing plug cavity primer percussion cavity, and face of flange	Visual
108. Diameter of firing plug cavity	Gage
109. Diameter of percussion primer large cavity	Gage
110. Diameter of percussion primer small cavity	Gage
111. Metal defective	Visual
112. Protective finish inadequate or defective	Visual
113. Burr	Visual
114. Foreign matter	Visual

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4.3.2.6 (Cont'd)

Defect		Method of Inspection
MINOR		
201.	Diameter of counterbore at flange end, min.	Gage
202.	Total diameter	Gage
203.	Depth of counterbore at flange end	Gage
204.	Depth from flange face to bottom of firing plug cavity	Gage
205.	Depth from flange face to bottom of larger percussion primer cavity	Gage
206.	Diameter of body	Gage
207.	Diameter at rear	Gage
208.	Width of undercut, rear of percussion primer small cavity	Gage
209.	Depth from rear face to bottom of igniter cavity	Gage
210.	Width of undercut, rear of igniter cavity	Gage
211.	Depth from rear face to start of igniter cavity	Gage
212.	Width of spring retaining cavity	Gage
213.	Width of flange	Gage
214.	Diameter of spring retaining cavity	Gage
215.	Chamfers and radius missing or incorrect	Visual
216.	Surface finish improper except as otherwise specified	Visual

4.3.2.7 Retaining Spring, Drawing B8593742 covering a detail of Drawing F11751078.

CRITICAL		
None defined		
MAJOR		
101.	Thickness of wire, min.	Gage
102.	Metal defective	Visual
103.	Protective finish inadequate or defective	Visual
104.	Burr	Visual
105.	Foreign matter	Visual
MINOR		
201.	Diameter, face	Gage
202.	Thickness of wire, max.	Gage
203.	Gap, min. (when spring is compressed to specified O.D.)	Gage

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4.3.2.8 Primer, Percussion, Drawing D10551686 covering a detail of Drawing F11751078.

Defect	Method of Inspection
CRITICAL	
1. Primer foil discolored	Visual
2. Mixed primers	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.3.2.9 Cartridge, Impulse CCU-28/A Assembly Drawing F11751078.

CRITICAL	
1. Improper assembly	X-ray 1/ - Visual
2. Any part missing	X-ray 1/ - Visual
3. Primer pellet broken, cracked or dislodged	X-ray 1/ - Visual
4. Metal defective	X-ray 1/ - Visual
5. Primer not properly seated	X-ray 1/ - Visual
6. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Crimp missing or improper	Visual
103. O-ring cut, split or cracked	Visual
104. Sealant missing	Visual
105. Sealant in cavity	Visual
106. Foreign matter	Visual
MINOR	
201. Diameter of body, max.	Gage
202. Diameter of head, max.	GAge
203. Total length, max.	Gage
204. Location of datum diameter	Gage
205. Varnish missing or inadequate	Visual
206. Fold, wrinkle or bulge	Visual

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4.3.2.9 (Cont'd)

1/ The cartridges shall be x-rayed with the sides of the case perpendicular to the rays. The x-rays negatives properly identified shall be forwarded to the procuring activity.

4.3.2.10 Box, Fiberboard (Inner) Packing, Ammunition for Cartridge, Impulse CCU-28/A Drawing D9282893.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual & Visual
202. Top of box loose	Manual
203. Strapping missng from box	Visual
204. Supports or fillers missing from box	Visual
205. Box damaged	Visual
206. Seal missing or improperly positioned	Visual
207. Marking missing, incorrect or illegible	Visual

4.3.2.11 Box, Packing Ammunition for Cartridge, Impulse CCU-28/A Drawing D9282894.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual & Visual
202. Hardware missing loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual

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4.3.2.11 (Cont'd)

Defect	Method of Inspection
MINOR (Cont'd)	
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provision of 4.4.

4.3.3.1 Vibration.- Twelve (12) of the twenty-one (21) x-rayed cartridges of the initial production sample, selected for ballistic tests and properly identified, shall be subjected to vibration as prescribed in 3.2. The twelve (12) samples for vibration shall be grouped and temperature conditioned, prior to vibration as indicated in Table IV below. After the vibration tests, the twelve (12) vibrated cartridges shall be x-rayed as directed in Note 1/ of 4.3.2.9. Evidence of any disintegration of the primer, propellant, ignitor, or derangement of any component shall be cause for rejection initial production sample. When a change occurs in the lot of primers, propellant, or igniters, after the initiation of production, 12 assembled cartridges from the first production lot shall be tested in accordance with this paragraph.

TABLE IV

Temperature	Not vibrated	Vibrated
-65°F to 70°F	3	4
65°F to 75°F	3	4
200°F to 205°F	3	4

4.3.3.2 Airtightness.-Each cartridge shall be subjected to the specified pressure differential. Failure to comply with the drawing requirements shall be cause for rejection of the cartridge.

4.3.3.3 Hardness.-Fifty (50) units from each heat treated lot of components shall be tested. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL-11751078.

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4.4 Test methods and procedures.

4.4.1 Vibration. The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.2 Airtightness.-The method for determining airtightness shall be by use of equipment listed on IEL-11751078.

4.4.3 Hardness.-A hardness testing machine shall be used for determining the hardness values specified on the applicable drawing.

4.4.4 Protective finish.-Test procedures for protective finish shall be in accordance with MIL-C-60539. The cartridge case and head shall be sectioned longitudinally to better expose the significant surfaces to the salt spray.

4.4.5 Sealant application.- The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length of the joining surfaces at the junction of the sealing disc and case sidewall and that the propellant cavity is free of sealant.

4.4.6 Functional.

4.4.6.1 Ballistics. The cartridge shall be loaded into an acceptable RAU-1/A Remover, divided into three (3) equal groups and temperature conditioned in accordance with Table I. The cartridge shall be tested within 5 minutes after removal from the conditioning chamber. When time is exceeded, the remover shall be reconditioned in accordance with 4.4.6.1.1.

4.4.6.1.1 Temperature conditioning.-A simulated RAU-1/A remover loaded with a dummy CCU-28/A cartridge, containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking.-Packing and marking shall be as specified by the procuring activity. Level A: Drawings D9282893 and D9282894. Level B - Drawings D9282893 and D9282895.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A567

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE :	1. DOCUMENT NUMBER MIL-C-48448(MU)	2. DOCUMENT DATE (YYMMDD) 740308

3. DOCUMENT TITLE

CARTRIDGE, IMPULSE, CCU-28/A ASSEMBLY

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
C. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial 301-744-1963 (2) DSN 354-1963
c. ADDRESS (*Include Zip Code*) Engineering Documentation (Code 4230) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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